EQAAB — عـقــآب

Intelligent Autonomous Aerial Surveillance System

by Abdulrahman Al-dahhas · Turki Al-otaibi · Saeed Al-harthi · Rayan Al-ghamdi · Hossain Al-hubailee · Mohammed Al-sharif · Omar Al-anzi
Supervisor: DR. Ahmed Bin Mahfoudh

TU جامعة الطائف TAIF UNIVERSITY
College of Computers & Information Technology
Taif University · KSA
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ABSTRACT

Wide-area security operations are constrained by limited coverage, delayed response, and reliance on human-operated systems. Eqaab presents an autonomous UAV-based surveillance platform that performs real-time onboard AI inference without cloud dependency or continuous operator control. The system employs an embedded ROS 2 pipeline on a Raspberry Pi 5 to enable fully self-contained perception, tracking, and alert generation. To overcome catastrophic forgetting in domain-specific detection, a dual-model YOLOv8 architecture is adopted, improving robustness across heterogeneous targets including persons, vehicles, and aerial threats. Real-time tracking and IFF classification are executed onboard, with structured alerts transmitted via 4G LTE to a ground control interface. The system achieves 94.3% mAP@50 at 8.3 FPS on resource-constrained hardware, with an end-to-end latency of 1.4 seconds, demonstrating a fully operational edge-AI surveillance platform suitable for real-world deployment.

PROBLEM STATEMENT

Border surveillance and critical infrastructure protection depend on systems that are increasingly inadequate for the scale and complexity of modern threats. Static camera networks create exploitable blind spots across wide terrain, while ground patrols introduce predictable coverage gaps limited by human endurance. Radar systems, although effective at detecting large aircraft and vehicles at range, consistently fail to detect low-altitude small UAVs — a growing threat vector — due to their limited radar cross-section and flight profile that falls below conventional radar sweep thresholds. When UAV platforms are deployed for surveillance, they typically require dedicated pilots and separate video monitoring operators — replicating the fatigue problem rather than solving it. The result is a persistent and dangerous capability gap: large threats are partially covered, small aerial threats go entirely undetected, and incidents are documented incompletely. In high-risk environments, an undetected drone is not a theoretical risk — it is an active vulnerability that existing infrastructure cannot close.

OBJECTIVES

1. **Autonomous Mission Execution**
   Design and implement a drone platform capable of executing complete surveillance missions — takeoff, waypoint patrol, target tracking, and return-to-home — without continuous operator input.
2. **Onboard Real-Time Detection**
   Deploy a dual-model YOLOv8 inference pipeline on the Raspberry Pi 5 to detect persons, vehicles, and hostile drones at operationally relevant frame rates, with no cloud dependency.
3. **IFF Classification**
   Implement an Identification Friend or Foe system that distinguishes authorised friendly aircraft from unknown aerial targets, defaulting to Unknown under all failure conditions.
4. **Georeferenced Alert Transmission**
   Fuse confirmed detections with real-time GPS coordinates and transmit structured alerts instantly over 4G LTE to a cloud backend.
5. **Live Ground Control Interface**
   Develop a browser-based GCS dashboard providing real-time drone position, detection alerts on an interactive map, and mission command capability.

METHODOLOGY

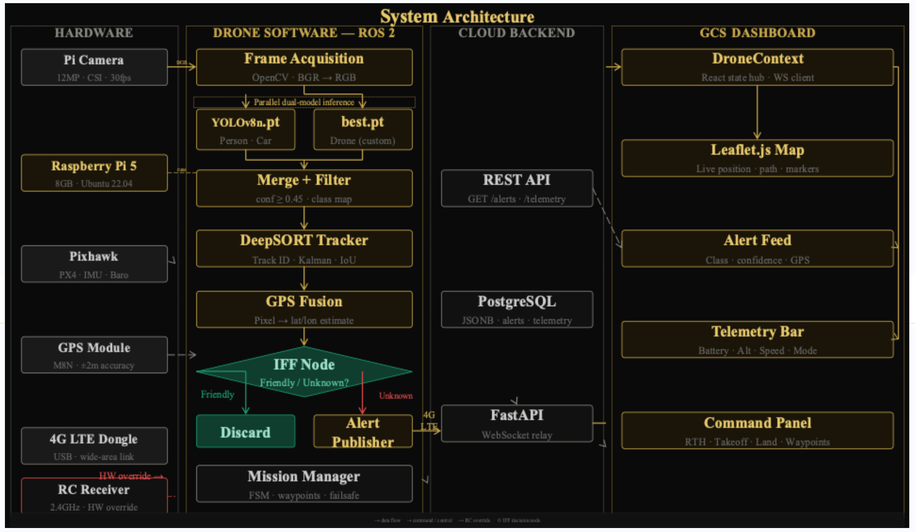
The Eqaab project follows an applied design-and-development approach, integrating aerial hardware with onboard AI to deliver an autonomous surveillance and interception drone for border security. Eqaab combines four subsystems: an F450 quadcopter platform controlled by a Pixhawk flight controller, a Raspberry Pi 5 onboard computer for AI inference, a YOLOv8-based detection and tracking module, and a ground monitoring interface. The Pixhawk and Raspberry Pi communicate via MAVLink, enabling autonomous flight commands driven by real-time vision data.

AI Pipeline. A custom dataset was annotated in YOLO format and split 70/20/10 for training, validation, and testing. Data augmentation (mosaic, flip, HSV jitter) was applied to improve robustness under varied lighting and altitudes. **YOLOv8n** was selected for its optimal speed–accuracy tradeoff on edge hardware, enabling real-time inference on the Raspberry Pi 5. The model was trained using the Ultralytics framework and optimized for deployment via NCNN. A lightweight tracking algorithm (ByteTrack) maintains target identity across frames, feeding coordinates to the interception logic.

Interception Logic. Detected target coordinates are translated into MAVLink flight commands, allowing the drone to autonomously align with and pursue the target without manual control.

Evaluation. System performance is measured using mAP@0.5, precision, recall, and inference speed (FPS) on the Raspberry Pi 5. Field tests assess detection accuracy across varying distances, lighting conditions, and target speeds.

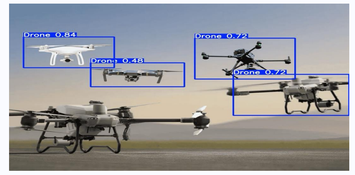
SYSTEM ARCHITECTURE



AI DETECTION PIPELINE & IFF

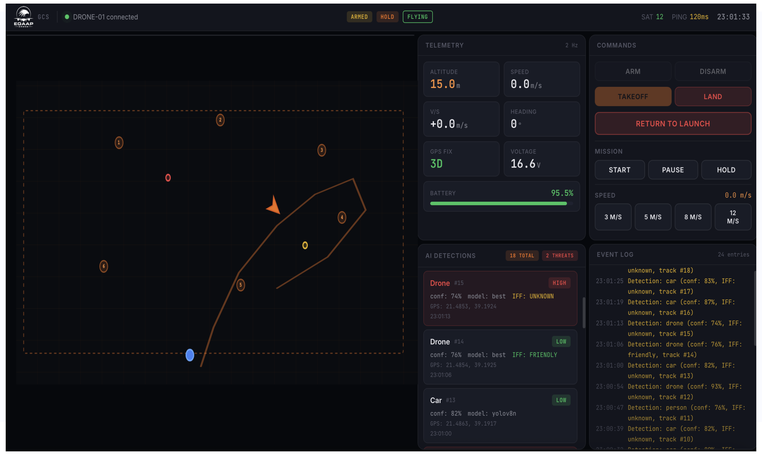
AI Detection Pipeline & IFF Classification The onboard pipeline runs two YOLOv8 models in parallel — yolov8n.pt for persons and vehicles, best.pt for drones — a parallel dual-model architecture adopted to avoid the catastrophic forgetting caused by single-model fine-tuning. Detections are merged, filtered at confidence ≥ 0.45, tracked via DeepSORT for persistent target identity, and fused with live GPS coordinates. Every detected drone is evaluated against an authorised IFF registry: matched targets are filtered out, while unmatched targets trigger an immediate alert. The system defaults to Unknown under all failure conditions, ensuring no threat is ever silently suppressed.

RESULTS



94.3%	8.3
mAP@50	*FPS on Pi 5*
1.4s	**≥0.45**
End-to-End Latency	*Confidence Threshold*

GCS DASHBOARD



CONCLUSION

Eqaab demonstrates that autonomous aerial surveillance with onboard AI threat classification is achievable on low-cost edge hardware. By combining a parallel dual-model YOLOv8 architecture, DeepSORT tracking, GPS fusion, and an IFF registry, the system distinguishes authorised assets from unknown drones in real time — addressing a critical gap in scalable border security. Its fail-secure design defaults every uncertain detection to **Unknown**, ensuring no threat is silently suppressed. Future work will extend the platform toward multi-drone coordination and integration with national surveillance infrastructure.

KEYWORDS

UAV · YOLOv8 · IFF · ROS 2 · Edge AI · DeepSORT · MAVLink · Surveillance

REFERENCES

Full references & repo